XIWEN CHEN

xiwenc@usc.edu | GitHub | LinkedIn | Personal Website | Open to Relocate

EDUCATION

University of Southern California — Los Angeles, CA
M.S. in Computer Science, Artificial Intelligence Track; GPA: 3.71 / 4.00 — *Jan. 2024 – May 2026*

South China Normal University — Guangzhou, China
B.E. in Software Engineering; GPA: 3.87 / 4.00 — *Sep. 2019 – Jul. 2023*

EXPERIENCE

American Best Life Inc. — Irvine, CA (Remote)
Software Engineer | Full-Stack & Agentic AI — *May. 2026 – Present*

- Owned **0-to-1 development and AWS deployment** of an AI wellness platform (Next.js, FastAPI, PostgreSQL, Terraform); a **1K-participant A/B test** improved completion rate from **43% to 78%** and cut average completion time by **45%**.
- Architected a **stateful LangGraph agent** combining **VLM-based label extraction**, **evidence-grounded RAG**, and constraint-aware recommendations; implemented versioned profile updates, optimistic concurrency, and idempotent async processing.
- Benchmarked the agent against a stateless LLM on 320 pilot recommendations, achieving **95%** constraint validity and **88%** expert approval; reduced end-to-end P95 latency by **27% (66s to 48s)** and inference cost per successful task by **35%**.

Keck School of Medicine, USC | Radiation Oncology — Los Angeles, CA
Research Lead | Agentic AI, Optimization & Evaluation | Advisor: Prof. Ruishan Liu — *May. 2025 – Oct. 2026*

- Led development of **RadOncGym**, a **Gymnasium-based agentic optimization framework** for radiotherapy planning, integrating a Gurobi-backed **12-variable multi-fidelity simulator** with modular agent interfaces and budget-aware execution.
- Built unified **evaluation infrastructure** for LLM/RL agents and numerical optimizers, integrating standardized interfaces, automated experiment orchestration, reproducible trajectory logging, and end-to-end evaluation.
- Benchmarked **3** frontier LLMs and **1** locally deployed open-source LLM against optimization baselines across **600 contexts**, reducing full evaluations by **81%** while retaining **95%** of objective gains under matched compute budgets.

2 Cube Global — Los Angeles, CA
AI Engineer Intern | ML Data Pipelines & Evaluation — *May. 2025 – Sep. 2025*

- Built Python/Librosa data curation and validation pipelines, processing **80+ hours** of paired audio and transcripts into **45K** aligned utterances and filtering **16%** misaligned samples for model adaptation and evaluation.
- Conducted **industry-focused ML R&D** for multilingual supply-chain TTS, adapting **F5-TTS** on curated domain data and benchmarking against zero-shot **Qwen3-TTS** on quality–cost trade-offs to inform technology selection.

Lab for ML, Health and Biomedicine, USC — Los Angeles, CA
Research Assistant | Data-Centric ML & Training Data Curation | Advisor: Prof. Ruishan Liu — *Sep. 2024 – May. 2025*

- Led **TAVO and TAC**, developing **budget-aware training data-curation** methods combining target-conditioned selection, multi-fidelity search, and active acquisition under distribution shift and constrained training budgets.
- Built reproducible **multi-GPU PyTorch/CUDA** training pipelines with **Slurm** orchestration and automated multi-budget benchmarking, validating TAVO across multi-backbones and CV tasks on BraTS 2021, MAMA-MIA, and OfficeHome.
- At a **12%** source-data budget, TAVO outperformed fixed-selector and domain-adaptation baselines by **+1.9/+2.2 Dice points**, while TAC exceeded the strongest baseline and full-source training by **+1.5/+4.1 points**, respectively.

Xiaomi Technology Co., Ltd. — Beijing & Shenzhen, China
Research Engineer Intern | Distributed Systems & Cloud Infrastructure — *May. 2024 – Sep. 2024*

- Integrated network- and erasure-coding components into the enterprise NCDSS **distributed-storage system**, deploying database, coordination, coding, and file-system services and validating K=4,M=2 erasure-coded storage across a **60-node** Linux environment.
- Built a layered reliability and failure-analysis workflow with **50+ client-I/O correctness scenarios** covering encoding, decoding, repair, update, end-to-end file I/O, and recovery under simulated storage failures across **2KB–1GB workloads**.
- Diagnosed system bottlenecks and benchmarked distributed coding implementations across **48 paired tests** and **8 baselines**, improving encoding throughput by **81.5%**, end-to-end upload throughput by **22%**, and recovery latency by **12%**.

TECHNICAL SKILLS

Languages: Python, C/C++, Java, SQL, Bash, TypeScript

Systems & Backend: Linux, Docker, Kubernetes, AWS, Terraform, CI/CD, FastAPI, PostgreSQL, Kafka, Spark, Distributed Systems

AI/ML Systems: PyTorch, CUDA, Slurm, Hugging Face Transformers, scikit-learn, Gymnasium, SGLang, LangGraph, RAG, Gurobi

SELECTED PUBLICATIONS & MANUSCRIPTS

RadOncGym: An Agent-Based Framework for Radiotherapy Dose-Mimicking Optimization [PDF] [Code] — WACV 2027
First Author | Agentic AI, LLM Agents, Planning & Evaluation, Multi-Fidelity Optimization — *Accepted*

TAVO: Target-Aware Source Curation for Budgeted Cross-Center Tumor Segmentation [PDF] [Code] — WACV 2027
First Author | Data Curation, Data-Efficient Learning, Domain Adaptation, Optimization — *Accepted*

Target-Anchored Coverage for Source–Target Curation in Cross-Center Medical Image Segmentation [Link] — ICLR 2027
First Author | Active Learning, Data Curation, Data-Efficient Learning, Domain Adaptation — *Under Review*

A Stroke Image Recognition Model Based on 3D ResNet and Attention Mechanism [Link] — JIFS 2022
Second Author | Deep Learning, 3D CNNs, Attention Mechanisms, Image Classification, DOI: 10.3233/JIFS-212511